CS331: Computer Networks

Project 02: Web Server Configuration in Cisco Packet Tracer

Total Weightage: 2.5 Marks — **Platform:** Cisco Packet Tracer

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1 Project Overview

The objective of this mini-project is to demonstrate practical knowledge of application-layer service deployments using Cisco Packet Tracer. Students will configure a dedicated Web Server hosting a customized Web Page and verify connectivity and HTTP service accessibility from network client end-devices.

2 Instructions & Guidelines

- **Collaboration:** This is a group assignment. Include all group member names and student IDs clearly in the web content as instructed.
- **Scope Limitation:** Configure **only** the Web Server (HTTP/HTTPS) and Host IP configurations. **Do NOT configure a DNS Server** for this task.
- **Submission:** Submit the completed Packet Tracer file (`.pkt`) along with a concise project report containing screenshots showing successful connectivity.
- **Reference Material:** Please find the detailed reference document and visual setup guidelines attached in the **Appendix** at the end of this document[Appendix].

3 Configuration Procedure

Follow these steps carefully to complete the project setup:

1. **Device IP Configuration:**
   - Click on the designated **Server** device[Appendix].
   - Navigate to the **Desktop** tab and select **IP Configuration**[Appendix].
   - Assign the required static **IP Address** and **Subnet Mask**[Appendix].
2. **HTTP Server Configuration:**
   - Navigate to the **Services** tab on the Server and select **HTTP**[Appendix].
   - Ensure both **HTTP** and **HTTPS** services are switched to **On**[Appendix].
   - In the File Manager list, **delete** all existing default HTML files (`copyrights.html`, `cscoplogo177x111.jpg`, `helloworld.html`, `image.html`) **EXCEPT** `index.html`[Appendix].
3. **Customizing HTML Content:**
   - Click **(edit)** next to `index.html`[Appendix].
   - Modify the HTML structure to display your group details as follows[Appendix]:

```
<html>
    <body>
        <h1>Welcome to our CS331 Project</h1>
        <p>[INSERT GROUP MEMBERS NAMES AND STUDENT IDs HERE]</p>
    </body>
</html>
```

- Save the changes to `index.html`[Appendix].

4. **Service Verification:**

- Open any client **Host** machine on the network[Appendix].
- Navigate to the **Desktop** tab and select **Web Browser**[Appendix].
- In the URL address field, enter the configured **IP Address** of the Web Server and click **Go**[Appendix].
- Verify that the custom HTML page displays your group members' details correctly[Appendix].

4 Evaluation Rubric (Total: 2.5 Marks)

The submission will be evaluated based on the criteria listed below:

Evaluation Criteria	Description	Marks
IP Setup & Addressing	Correct static IP address and subnet mask assignment on both Server and Client devices.	0.5
HTTP Service Setup	Correct enablement of HTTP services and clean cleanup of default files.	0.5
HTML Modification	Proper modification of `index.html` displaying all required group details clearly.	0.5
Functional Verification	Successful retrieval and rendering of the page via Host Web Browser using Server IP.	0.5
Documentation & File	Valid `.pkt` file submission with complete verification screenshots.	0.5
Total		**2.5 Marks**

Table 1: Mini-Project Grading Rubric

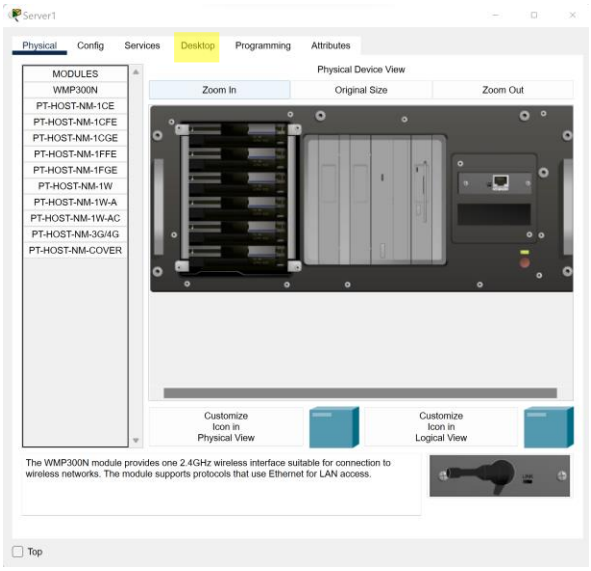
Appendix: Reference Setup Document

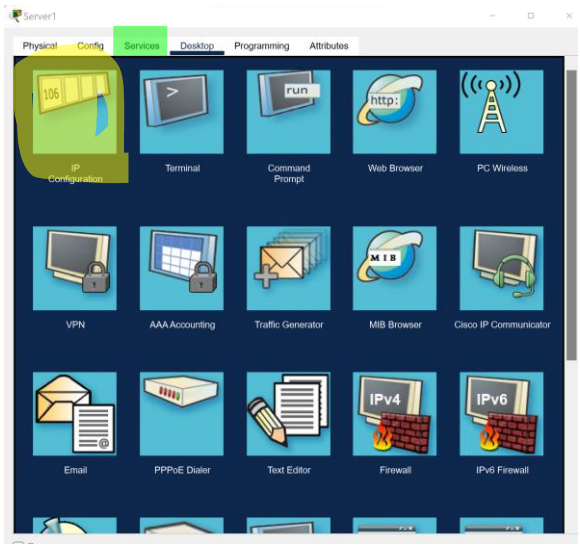
Please find the reference visual guide and original setup instructions attached below[Appendix].

[Attached reference document content / screenshots go here]

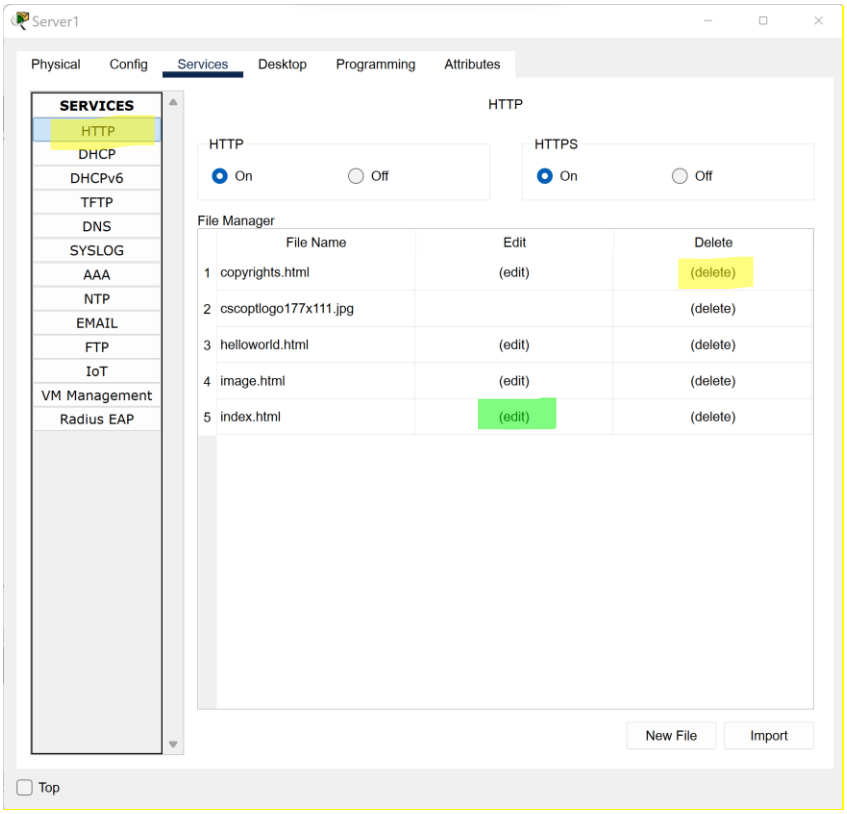
Server Configuration steps

1- Click on the Server
2- Select Desktop

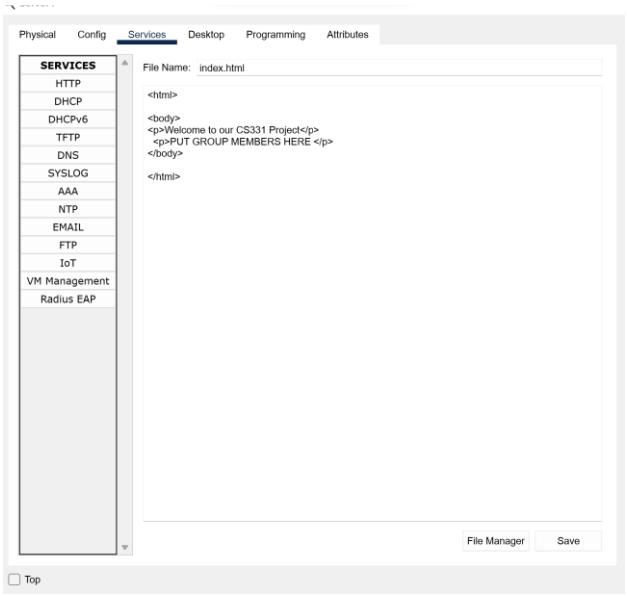




3- Select IP Configuration
4- Enter the IP address and the subnet mask
5- To configure the HTTP server, go to services then click on HTTP

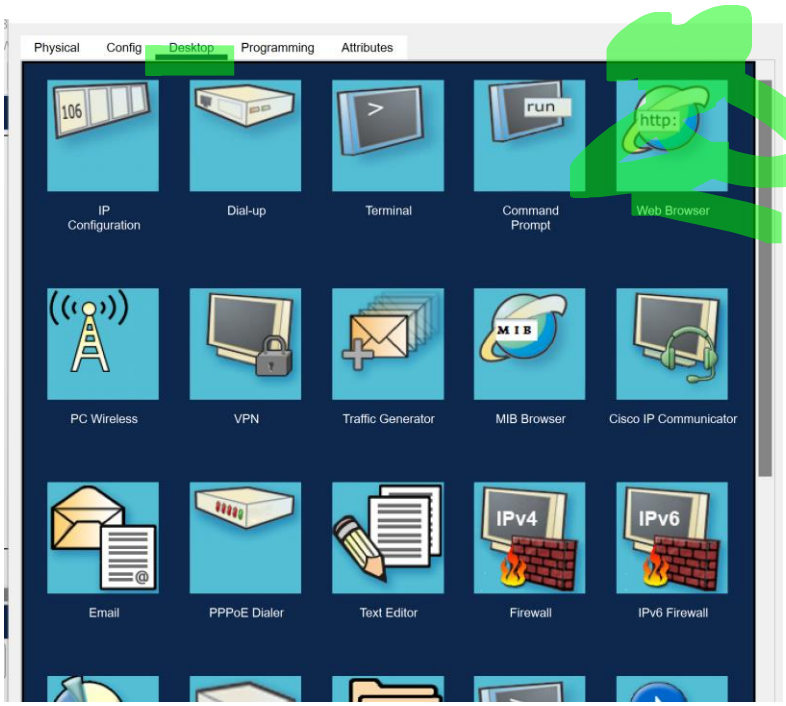


6- Then delete all of the files except the index.html and edit it.
7- Edit index.html

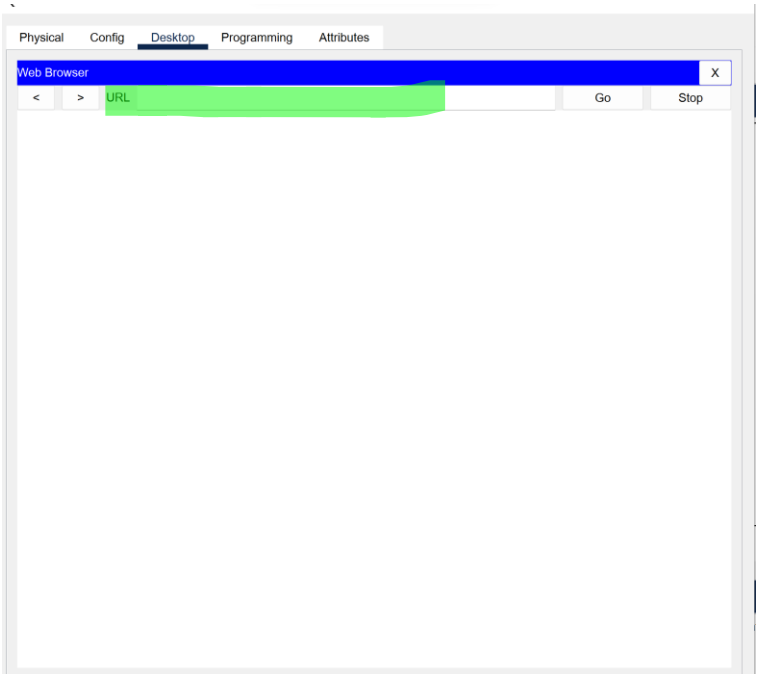


8- Verify the server by using the web browser in one Host.

Click on one host – select Desktop then Web Browser



9- Enter the IP address of the web server and click on GO.



It will show the results.

You can follow the steps in this video (BUT DON'T CONFIGURE THE DNS SERVER)